ACADEMIC USE DISCLAIMER:

THIS DESIGN IS FOR ACADEMIC AND INSTRUCTIONAL USE AT CSULB ONLY.
IT HAS NOT UNDERGONE REGULATORY TESTING OR SAFETY CERTIFICATION OF ANY KIND.
NOT INTENDED FOR COMMERCIAL DEPLOYMENT OR ANY OPERATIONAL UAV APPLICATIONS.
DO NOT USE IN SYSTEMS WHERE FAILURE MAY CAUSE INJURY OR DAMAGE.
CSULB AND THE DESIGNER DISCLAIM ALL LIABILITY FOR ANY MISUSE OR HARM.
NO WARRANTIES, EXPRESS OR IMPLIED, ARE PROVIDED WITH THIS HARDWARE.
ALL USE, MODIFICATION, OR ASSEMBLY IS ENTIRELY AT THE USER'S OWN RISK.

UNAUTHORIZED COPYING, DISTRIBUTION, OR REPRODUCTION OF THIS DESIGN IS PROHIBITED.
THIS MATERIAL IS FOR ACADEMIC USE AT CSULB AND NOT COMMERCIAL PURPOSES.
DO NOT INCLUDE THIS DESIGN IN ANY PRODUCT INTENDED FOR SALE.
MODIFICATION OR REDISTRIBUTION WITHOUT PERMISSION IS AT THE USER'S RISK.
CSULB AND THE DESIGNER DISCLAIM ALL LIABILITY FOR RESULTING DAMAGE OR LOSS.

1. INTERPRET DRAWING IAW STANDARDS PRESCRIBED BY ASME Y14.100.
2. UNLESS OTHERWISE SPECIFIED:

   ALL RESISTORS ARE 1%, 0.1W, 0603

   ALL CAPACITORS ARE 10%, 16V, 0603
3. PARTIAL REFERENCE DESIGNATIONS ARE SHOWN FOR COMPLETE.
4. SEE TABLE A FOR POWER AND GROUND INPUTS TO MICROCIRCUITS.
5. INTERPRET REFERENCE DESIGNATIONS AS FOLLOWS:

   EXAMPLE: U1 - 2

   U1 = COMPONENT; 2 = PIN NUMBER
6. GENERIC PART NUMBERS ARE SHOWN FOR REFERENCE ONLY.
7. CONNECT CHASSIS TO THE MOUNTING HOLES / CHASSIS GROUND.
8. LATEST REVISION CHANGES: NaN
9. UNUSED REFERENCES: R40, R41, R42, C28, C29, C30, C38, C39, C40

   → DO NOT INSTALL THESE COMPONENTS. LEAVE THEM UNPOPULATED.

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.
	-	RELEASED PER ENGINEERING CSULB-DEESC	260207	

TABLE A

REFERENCE DESIGNATION	POWER INPUT TO PINS							
	PWR	SRC						
	+24 V	6S BATT						
	+10 V	PSU						
	+3.3 V	BUCK						
	GND	RTN						

INSTALLATION GUIDE:

1. ESC BOARD GOES BELOW THE MCU BOARD. INSTALL NORMAL 1X12 FEMALE HEADER PINS INTO P2 AND P3.
2. A SWITCH AND APPROPRIATE FUSE SHALL BE SOLDERED TO THE BATTERIES POSITIVE WIRE. THE SWITCH ACTS AS MASTER POWER. REFER TO LITTLEFUSE FUSE FACTS AND FUSE SELECTION GUIDE TO DETERMINE THE APPROPRIATE FUSE.
3. AFTER FUSE AND SWITCH IS INSTALLED, SOLDER THE BATTERY/POWER SOURCE WIRES TO THE PADS LABELED BATT+ AND BATT-. BE CAREFUL TO NOT MIX THESE UP AS THERE IS NO REVERSE POLARITY PROTECTION.
4. C1 AND C2 SHALL BE INSTALLED ONTO THE BOTTOM LAYER. THE 3D HOUSING FOR THE ESC WILL BE DESIGNED TO HIDE THESE CAPS.
5. INSTALL 1X4 HEADER PINS INTO P4. BE CAREFUL TO INSTALL HEADER PINS WITH OR WITHOUT EXTENDED PINS DEPENDING ON THE STACK CONFIGURATION. THE ESC BOARD IS DESIGNED TO GO BELOW THE MCU BOARD.
6. MOUNTING HOLES ARE SIZED FOR M3 SCREWS. MOUNTING HOLES ARE NOT PLATED WITH COPPER.
7. THIS BOARD SUPPORTS AN EXTERNAL BUCK AT P1. REFER TO P1 ON THE BOARD FOR THE CORRECT PINS FOR THIS INSTALLATION.
8. DNI R1,R2,R3 IF THE DRV8323 IS CONFIGURED FOR SINGLE (1X) PWM MODE.
   FOR MCU BOARD ONLY: DNI R14, R15, R16, C22, C23, C24 IF THE DRV (6X) PWM MODE.
   DO NOT USE EXTERNAL BUCK WITHOUT DISABLING DRV8323 BUILT-IN BUCK CONVERTER.
   REMOVE R12 IN THE CASE THAT A EXTERNAL BUCK CONVERTER IS USED.
   DNI C17 IF THE DRV8323 AMPLIFIERS ARE CONFIGURED FOR MOSFET VDS CURRENT SENSING.
   DNI R9, R10, R11 IF THE DRV8323 AMPLIFIERS ARE CONFIGURED FOR DC BUS CURRENT SENSING.
   DNI R27, R29, R32 IF THE DRV8323 IS CONFIGURED FOR SINGLE (1X) PWM MODE.
9. THE SOLDERABLE PADS FOR THE MOTOR ARE LABELED WITH THEIR RESPECTIVE PHASES (A, B, AND C).
10. IF MOUNTING TO A 3D-PRINTED HOUSING, TAKE INTO CONSIDERATION THE COMPONENTS INSTALLED ONTO THE BOTTOM LAYER. USE SCREW SPACERS/STANDOFFS TO NOT DAMAGE THESE COMPONENTS.

NOTICE OF OWNERSHIP AND INTELLECTUAL PROPERTY

THIS SENIOR DESIGN PROJECT IS INDEPENDENTLY CONCEIVED, DESIGNED, AND FUNDED SOLELY BY THE PROJECT TEAM. NO FINANCIAL SUPPORT, SPONSORED RESEARCH, OR PROPRIETARY RESOURCES WERE PROVIDED BY CSULB OR ITS AFFILIATES. ALL DESIGNS, HARDWARE, SOFTWARE, AND DOCUMENTATION ARE THE EXCLUSIVE PROPERTY OF THE PROJECT TEAM. CSULB CLAIMS NO OWNERSHIP, LICENSING, OR COMMERCIAL RIGHTS EXCEPT FOR ACADEMIC EVALUATION PURPOSES.

CONTRACT NO	EE400D	
DWN BY		
CHECK BY	J TOLDEN	
APPV BY	J TOLDEN	
PROF	DR. JUFOUT	
MKS Rev MKS Date	JON TOLDEN, BAO PHAM, AUTUMN AGUILAR	

DEESC

CALIFORNIA STATE UNIVERSITY - LONG BEACH
ELECTRICAL ENGINEERING
LONG BEACH, CALIFORNIA

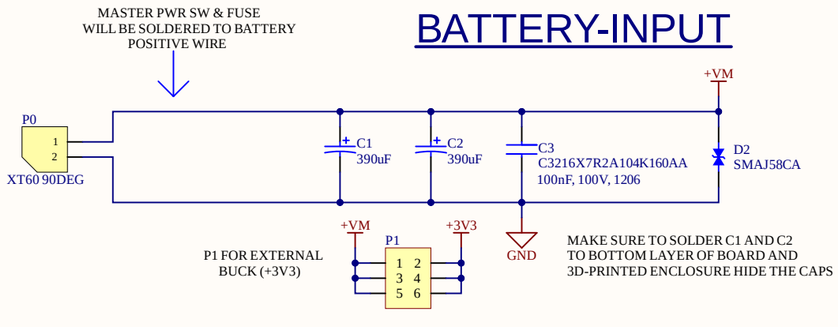
TITLE

MCU CONTROL FOR ESC

SIZE		DWG NO	REV
B	SPRING 2026 CSULB EE400D SENIOR DESIGN	DRV_1	-
SCALE: NONE	VIEWDRAW	SHEET 1 OF 2	

BLDC MOTOR CONTROL

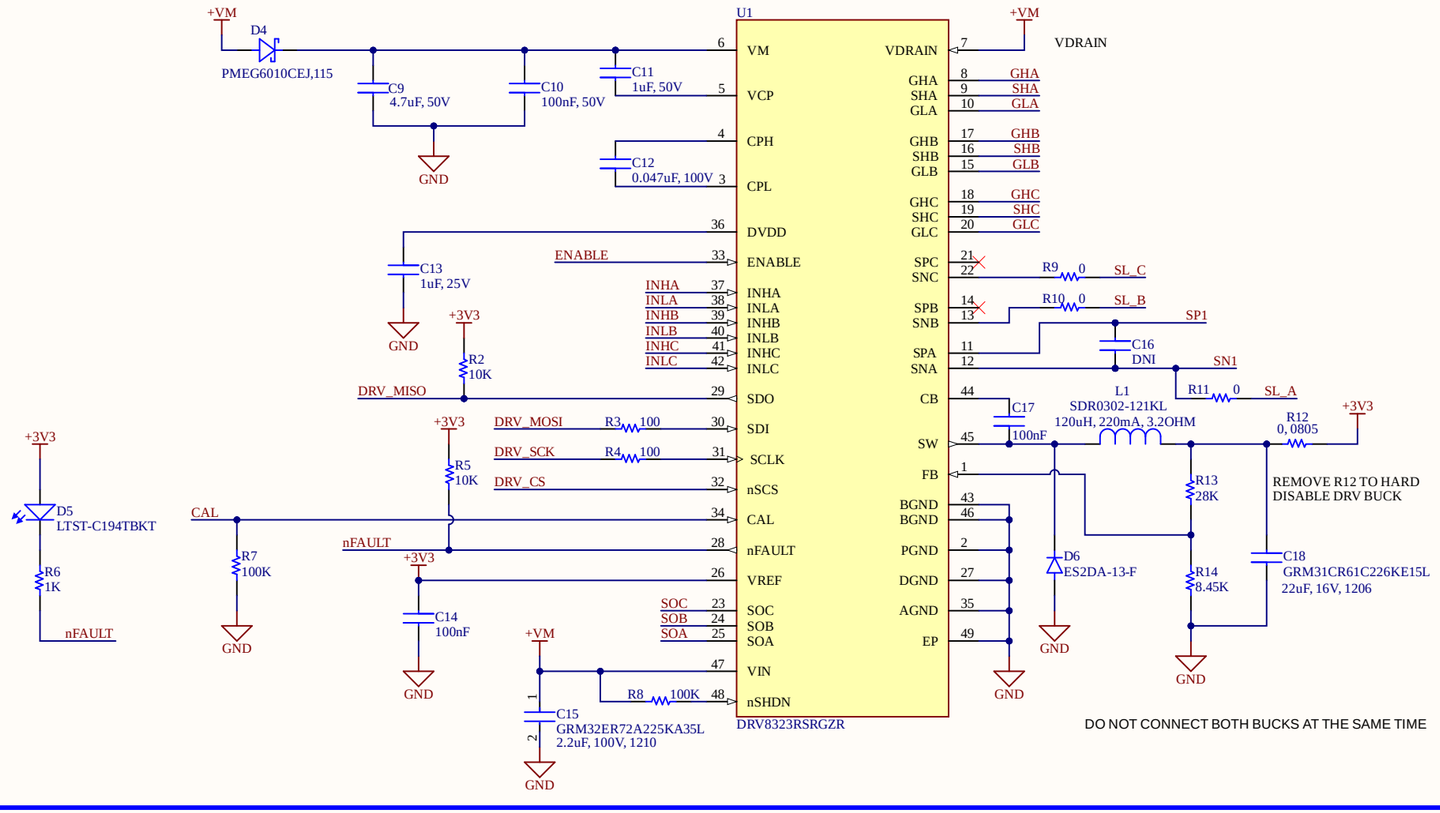
BATTERY-INPUT



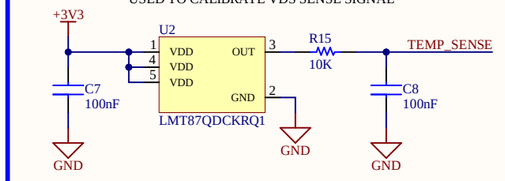
TEST POINTS



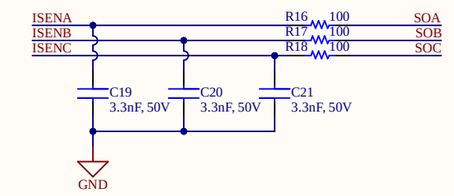
BLDC MOTOR DRIVER



TEMP SENSOR

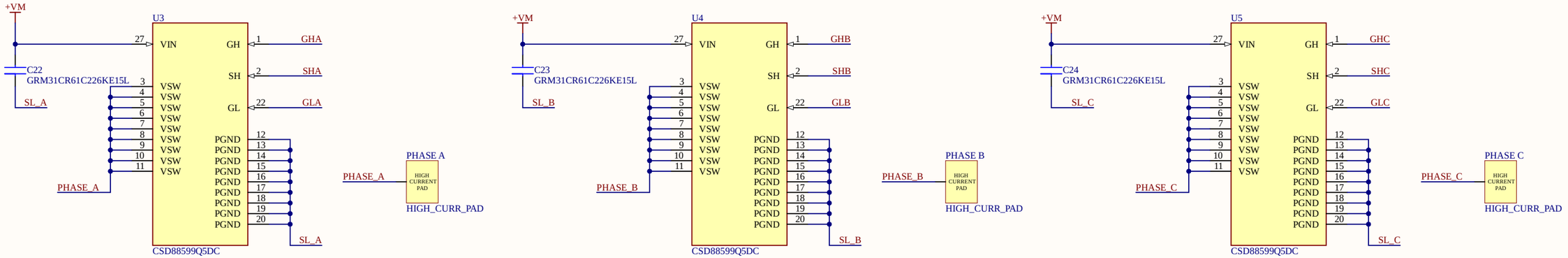


CURRENT SENSE

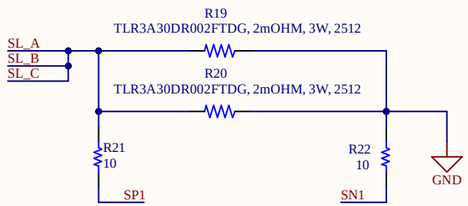


Note: No need to populate C17, if the DRV8323 amplifiers are configured for MOSFET VDS current sensing
Do not populate R9, R10, R11 if the DRV8323 amplifiers are configured for DC bus current sensing using the shunt resistor
Do not populate R27, R29, R32 if the DRV8323 is configured for single (1x) PWM

HALF BRIDGES

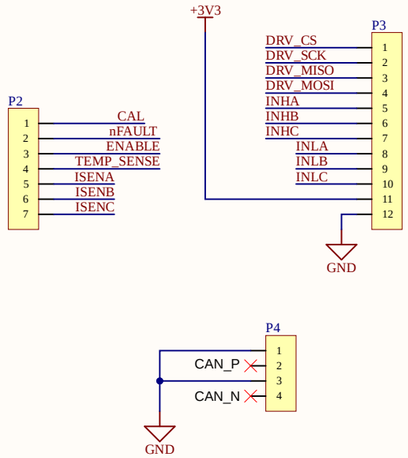


CURR SENSE



DNI R21 & R22 IF THE DRV8323 AMPLIFIERS
ARE CONFIGURED FOR MOSFET VDS CURRENT SENSING

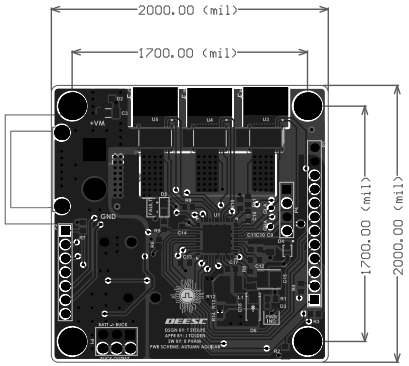
INTERFACE

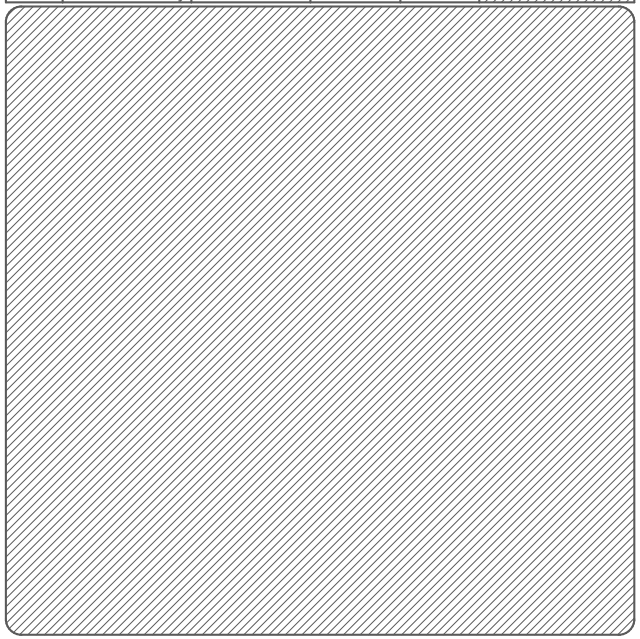


Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.40mil	3.5	
1	LAYER 1		1.40mil		
	Dielectric 2	PP-006	2.80mil	4.1	
2	LAYER 2	CF-004	1.38mil		
	Dielectric 1	FR-4	12.60mil	4.8	
3	LAYER 3	CF-004	1.38mil		
	Dielectric 3	PP-006	2.80mil	4.1	
4	LAYER 4		1.40mil		
	Bottom Solder	Solder Resist	0.40mil	3.5	
	Bottom Overlay				



LAYER 1: SIGNAL
LAYER 2: GND
LAYER 3: +3.3 V
LAYER 4: SIGNAL





DESIGN INFORMATION

MIN. TRACK WIDTH: 10 MIL
MIN. CLEARANCE: 6 MIL
MIN. VIA PAD SIZE: 20 MIL
MINIMUM ANNULAR RING 0.05mm (2MIL) EXTERNAL
PER IPC-D-275 CLASS 2 LEVEL C
REGISTRATION TOLERANCES: METAL +/- 5 MIL, HOLES +/- 3 MIL
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED): +/- 3 MIL

MATERIAL:
[ ] FR-408 [X] FR-4 High Tg [ ] OTHER
THICKNESS: [X] 62 MIL (1.6mm) +/-10% [ ] OTHER
TOLERANCE: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-
BOW & TWIST: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-

DRILLING:
REFERENCE: [X] AS SHOWN [X] NC_DRILL FILES
PTH COPPER THICKNESS: [X] 20-30 um [ ] OTHER

BOARD FINISH:
SILKSCREEN: [X] TOP [X] BOTTOM
SILKSCREEN COLOR: [X] WHITE [ ] OTHER
SOLDER RESIST COLOR: [ ] GREEN [X] OTHER RED
[X] MATTE [ ] SEMI-GLOSS
SURFACE FINISH: [X] IMMERSION GOLD (ENIG) [ ] ENEPIG
[ ] IMM. TIN/SILVER OR EQUIV [ ] OTHER
ARRAY/PANEL: [ ] CUT AND TRIM PER M1 BOARD OUTLINE
[ ] N.C. ROUTE [X] V. SCORE
CERTIFICATION: MATERIALS AND WORKMANSHIP FOR ALL PCBS TO MEET OR EXCEED THE REQUIREMENTS OF:
[X] ANSI IPC-A-600F CLASS -> [ ] 1 [X] 2 [ ] 3
[X] RoHS [ ] OTHER PER ORDER
ALL BOARDS MUST MEET OR EXCEED EE400D REQUIREMENTS.

ADDITIONAL REQUIREMENTS:
MICROSECTION: [ ] YES
BARE BOARD ELEC. TEST: [ ] NONE [X] REQUIRED [ ] PER ORDER

1000.00mil

COMPONENTS MARKED 'DNI' SHOULD NOT BE POPULATED.
ASSEMBLY VARIANT: [No Variations]

DEESC

TEAM LEAD: JON TOLDEN
THOMAS STOLPE
BAO PHAM
AUTUMN AGUILAR

PROJECT TITLE: ESC PCB FOR BLDC MOTOR
DESIGNED FOR: CSULB EE400D SENIOR PROJECT
FILE NAME: ESC_EE400D_PCB.PcbDoc

PCB VIEWED FROM TOP SIDE		
SOFTWARE BY: BAO PHAM		
POWER SCHEME BY: AUTUMN AGUILAR	GENERATED : 8/7/2026 8:51:05 PM	

NOTICE OF OWNERSHIP AND INTELLECTUAL PROPERTY:
THIS SENIOR DESIGN PROJECT IS INDEPENDENTLY CONCEIVED, DESIGNED, AND FUNDED ENTIRELY BY THE PROJECT TEAM. NO FINANCIAL SUPPORT, PROPRIETARY MATERIALS, OR DIRECTED RESEARCH WERE PROVIDED BY CSULB OR ITS AFFILIATES. ALL INTELLECTUAL PROPERTY, DESIGNS, HARDWARE, SOFTWARE, AND DOCUMENTATION REMAIN THE SOLE PROPERTY OF THE PROJECT TEAM. CSULB CLAIMS NO OWNERSHIP, LICENSING, OR COMMERCIAL RIGHTS BEYOND ACADEMIC EVALUATION PURPOSES.

PROJECT MANAGER: JON TOLDEN	LAYOUT BY: THOMAS STOLPE
SCALE: 0.72	ALTIUM DESIGNER VERSION: 26.8.1.31